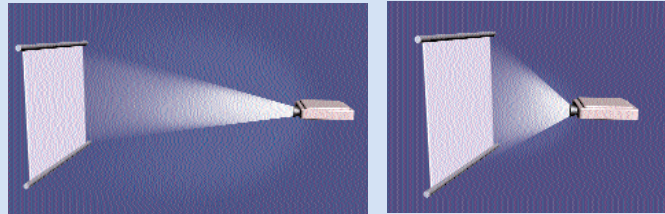


## The Lens Of The Matter: Long And Short Throw

There are mainly two types of lenses for projectors, based on the relationship between the size of the image projected and the distance between the projector and screen. A general rule of thumb for a standard projector lens is a foot of screen for every two feet of distance between the projector and the screen. Long-throw lenses will be able to maintain a larger distance between the screen and the projector while providing a clear image of the same size. Similarly, a short-throw lens will be able to provide the same image size while maintaining minimal distances between the projector and screen.

Most projectors ship with a default lens that is neither truly long throw nor short throw, but something in between—so that they're suitable for a larger range of situations. If you wish, you can always change the lens—for example, if you're only going to use the projector in a large hall, get yourself a long throw lens.

Comparing the two types of lens is impossible—their suitability depends on the conditions—namely the distance between the projector and screen. For large halls where the projector may be set up very far from the screen, obviously a long-throw lens would be preferred. Not only will the image size be controllable, such lens can also project a clear image even over relatively large distances.



A long throw lens at work on the left, short throw on the right. Notice the distance between the lens and the screen in each case.

A short-throw lens would project an image that would overflow the screen; moreover, the image would be unfocused.

A short-throw lens is suitable for small conference rooms or classrooms—basically any place where the distance between projector and screen is insubstantial. A long-throw lens would project too small an image if used in such a situation.

Nearly all projectors (standard, long-throw, and short throw) sport zoom lenses. This allows them to compensate somewhat for the inherent shortcomings in the lens type used.

## ZEN PTV-01B

**A bonsai, this...**

The PTV-01B was the dark horse in our tests, a new brand in the Indian projector market. At a glance, it looks like a huge toy, with body plastic reminiscent of Leo-Mattel's toys. That's not a bad thing, and it feels sturdy to hold. If you can hold it, that is—at a shade less than 6 kg, it's heavy! It's by far the bulkiest projector we received.

This is a strictly a home-entertainment-only projector, and the PTV-01B makes no bones about this, losing out on USB and wireless connectivity. It also does without any DVI inputs. Zen assumes you have a discrete sound setup as well, so the PTV-01B does without any audio inputs. There's a cable TV input though, something we didn't see on any of the other contenders. It gives a whole new meaning to the term "large-screen TV"!

We quickly ran into problems setting this one up. The PTV-01B doesn't have any zoom capabilities—this is absolutely shocking! The focus is adjustable (though it isn't very effective). The only way to zoom in or out is to actually move the projector closer or further from the screen.

The DisplayMate tests weren't completed to our satisfaction. In fact, calibrating the PTV-01B with a setup menu devoid of too many options was difficult. The



Sharpness and Resolution tests saw severe problems with the horizontal and vertical bar resolutions. The PTV-01B just could not render this test satisfactorily. This was also the only projector where there were noticeable problems with the corner resolution test.

Moving on to the colour tests, we saw problems with nearly all the tests, with clear anomalies visible on-screen.

The Zen PTV-01B isn't suitable for presentation work; text was unreadable, and we developed headaches just straining our eyes to make out what was written on-screen. Our PowerPoint designs and photograph tests fared no better. To Zen's defence, the PTV-01B isn't aimed at such applications. Let's go on to see how it performed in its area of competence.

Although colours were rendered as crisply as we'd like, the PTV-01B tried its best to perform at least decently here. It managed to do so, but with some very mediocre scores. The BenQ MP611c makes a much better buy

at the same price point if movie-watching is your thing.

Do to its poor contrast ratio and low sharpness, the Zen PTV-01B isn't a gamer's dream—"nightmare" is more like it, as we found when playing F.E.A.R.! At Rs 42,000, the only thing it has going for it is its price. Not a big deal, since the BenQ MP611c swings into the same budget range.

### What we think

The first difference noticeable from last year's comparison is the ratio of XGA to SVGA projectors we received. This year we received seven XGA projectors and just two SVGA projectors. This shows a trend that manufacturers are targeting the Indian market for higher-end products. This is partially due to the increase in the demand for projectors, and the penetration of these devices into the domestic arena.

Surprisingly, the increase in demand has nothing to do with falling prices, and price reductions, if any, will arise from further increase in demand—a most uncommon phenomenon in India, which is primarily a value-oriented market.

A growing number of corporate customers have adopted projectors for all their boardroom and sales pitch needs. A much smaller fraction of home users have also populated the projector bandwagon. We're hoping to see price reductions next year! ■

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